SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	Harshavardhan RV	**Reg. No:**	192472163
Section / Batch:	CSE-AI	**Date:**	8/8/2026

Code of Conduct

I __HARSHAVARDHAN. RV(192472163)______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Harshavardhan RV_______________________

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- λ_1 = 0.5 (Trigram)
- λ_2 = 0.3 (Bigram)
- λ_3 = 0.2 (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

CASE STUDY 1 – Smart Mobile Keyboard Prediction System

Aim: To analyze Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy for next-word prediction in a smart keyboard.

Q1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability . Interpret its role.

Formul

a Given

-
-

Calculation

Answer

Bigram Probability =

1.0 Interpretation

A probability of **1.0 (100%)** means that every occurrence of the word **"data"** in the corpus is followed by **"science"**. Therefore, the keyboard confidently predicts **science** whenever the user types **data**, improving prediction accuracy and typing speed.

Q2. Apply the Backoff Model for the unseen sequence "data science improves". Solution

The trigram **"data science improves"** does **not appear** in the training corpus. Using the **Backoff Model**:

1. Check Trigram → **Not Found**
2. Backoff to Bigram → **science improves** → Not Found
3. Backoff to Unigram → **improves**

Since only the unigram is available,

Answer

The probability is estimated using the **lower-order unigram probability**.

Explanation

The Backoff Model avoids assigning zero probability to unseen word sequences. This enables predictive keyboards to continue suggesting reasonable words even when users type new or rare phrases, making real-time prediction more robust.

Q3. Compute Deleted Interpolation Probability for "data science is".

Formula

Given

-
-
-

Using the corpus:

- Trigram probability = 1
- Bigram probability = 1
- Unigram probability = 1

Calculation
Answer

Deleted Interpolation Probability =
1.0 Discussion

Deleted Interpolation combines trigram, bigram, and unigram probabilities. This balances high accuracy from higher-order models with better coverage from lower-order models, reducing problems caused by sparse data.

Q4. Calculate the Entropy.

Given

-

Entropy = 0.92
bits

Interpretation

The entropy is relatively low, indicating **less uncertainty** in predicting the next word. Since **"is"** has a much higher probability than **"drives"**, the keyboard predicts **"is"** with greater confidence, improving prediction reliability and user experience.

Case Study 1 Result

The Bigram, Backoff, Deleted Interpolation, and Entropy techniques together improve next-word prediction accuracy by handling both common and unseen word sequences efficiently in AI-powered mobile keyboards.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 2: AI-Powered Customer Support Chatbot

1. Penn Treebank POS tagging of both sentences
The Penn Treebank (PTB) tagset assigns grammatical categories such as **VB (verb, base form)**, **NN (noun, singular/common noun)**, **DT (determiner)**, **JJ (adjective)**, and **RB (adverb)**.

Sentence 1: "Book a flight ticket now."

Word	POS Tag	Explanation
Book	**VB**	Base-form verb; it is an instruction/command to make a reservation.
a	**DT**	Determiner introducing the noun phrase "flight ticket."
flight	**NN**	Noun describing the type of ticket.
ticket	**NN**	Common noun; object of the verb "Book."
now	**RB**	Adverb indicating when the action should occur.

Tags: Book/VB a/DT flight/NN ticket/NN now/RB
Here, **"book" is a verb** because the sentence is an imperative command. The implied subject is "you": **"[You] book a flight ticket now."** Therefore, *book* represents an action.

Sentence 2: “This book is interesting.”

Word	POS Tag	Explanation
This	**DT**	Determiner modifying “book.”
book	**NN**	Singular common noun; it refers to a physical/publication object.
is	**VBZ**	Third-person singular present form of the verb “be.”
interesting	**JJ**	Adjective describing the book.

Tags: This/DT book/NN is/VBZ interesting/JJ
Here, **“book” is a noun** because it is the subject of the sentence and is modified by the determiner **“This.”** The phrase “is interesting” also describes the book as an object.

---

2. HMM probability for “book” as a Verb
The given probabilities are:
For the first word, the HMM can calculate the score for each possible tag as:
For the noun:
Therefore:

Tag	Start Probability	Word Probability	HMM Score
VB	0.5	0.6	**0.30**
NN	0.5	0.4	**0.20**

The HMM therefore **favors VB**, because:
The relative likelihood of VB versus NN is:
Thus, based only on the supplied probabilities, **“book” is 1.5 times more likely to be tagged as VB than NN**.
The probability can also be normalized:
So the model assigns a **60% probability to VB** and **40% to NN**, given the provided probabilities.
In the actual sentence, the contextual clue **“a flight ticket”** further supports the verb interpretation: *Book a flight ticket* is an action/command.

---

3. Rule-Based Tagging vs. Stochastic (HMM) Tagging

Aspect	Rule-Based Tagging	Stochastic/HMM Tagging
Approach	Uses manually created linguistic rules	Uses probabilities learned from data
Context	Depends on explicitly written rules	Uses probabilities of tags and word/tag sequences
Ambiguity	Can resolve common ambiguities with appropriate rules	Handles ambiguity probabilistically
Scalability	Difficult when rules become numerous	More scalable with sufficient training data
Adaptability	Requires manual modification of rules	Can be retrained using new data
Maintenance	Rules can become complex	Model can be updated with new training data
Example	“If *book* occurs before a determiner, tag it as VB”	Compare (P(VB

Rule-based tagging can be accurate for predictable grammatical patterns. For example, a rule can identify *book* as a verb when it occurs at the beginning of an imperative sentence.
However, a large airline chatbot encounters huge numbers of different expressions, abbreviations, spelling variations, and ambiguous words. Creating and maintaining rules for every situation would be difficult.
HMM tagging is generally more suitable for large-scale deployment because it can learn statistical patterns from large amounts of tagged language data. It can use both lexical probabilities and contextual/tag-transition probabilities to choose the most likely POS tag.
Therefore, **a stochastic approach is preferable for large-scale chatbot deployment**, although a practical system can combine statistical models with linguistic or business-specific rules for improved accuracy.

---

4. Role of standardized POS tagsets and word classes in chatbot performance
A standardized POS tagset such as the **Penn Treebank tagset** provides a consistent representation of grammatical information. This is important because the same word can have different meanings and grammatical roles depending on its context.
For example:

a. Improving intent detection
POS information helps the chatbot identify what the user wants.
For example:
"Book a flight to Delhi."
The verb **"Book/VB"** indicates an **action/request**, while nouns such as **"flight"** provide information about the object of the request.
This can help classify the user's intent as **flight booking** rather than, for example, a question about a book.

b. Improving response generation
POS tags help the system understand relationships between words and generate more appropriate responses.
For example:
"Cancel my flight."
The system can recognize **"Cancel" as a verb/action** and **"flight" as a noun/object**, making it easier to map the request to a flight-cancellation operation.

c. Handling lexical ambiguity
Words such as *book*, *check*, *fly*, and *change* can have multiple grammatical roles.
POS tagging allows the chatbot to distinguish their usage based on context:

- **"Book my ticket." → book/VB**
- **"I bought a book." → book/NN**

This reduces misunderstanding and improves intent classification.

d. Improving overall chatbot performance
Standardized word classes provide structured linguistic information that can be used by downstream NLP components. They can improve:

- **Intent classification**
- **Entity and information extraction**
- **Semantic interpretation**
- **Dialogue management**
- **Response generation**
- **Context understanding**
- **Handling ambiguous words**

Conclusion
POS tagging is an important component of an AI-powered airline chatbot. In the first sentence, **"book" is VB** because it expresses an action in an imperative sentence, while in the second sentence it is **NN** because it refers to a publication/object. Based on the supplied HMM probabilities, the VB score is **0.30**, compared with **0.20 for NN**, so the model favors **VB**. For large-scale chatbot systems, **stochastic/HMM-based tagging is generally more flexible and scalable than purely rule-based tagging**, while standardized POS tags provide a consistent linguistic foundation for accurate intent detection and response generation.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    - economic (120 occurrences)
    - growth (450 occurrences)
    - increases (210 occurrences)
    - employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

CASE STUDY 3: News Analytics and POS Tag Correction System

1. Applying the Transformation-Based Rule

The sentence is:
"Economic growth increases employment."
Initial POS tags:

- **economic/JJ**
- **growth/NN**
- **increases/NNS**
- **employment/NN**

The learned rule is:

Change NNS to VBZ if the preceding word is tagged as NN.
The word “increases” is currently tagged NNS. Its preceding word, “growth,” is tagged NN. Therefore, the rule applies:

Corrected POS sequence
economic/JJ growth/NN increases/VBZ employment/NN
The rule is linguistically appropriate because “increases” functions as the main verb of the sentence. The subject is “economic growth,” which is singular, so the verb should be in the third-person singular present form “increases” (VBZ).
The sentence can be understood as:
Economic growth → increases → employment

Thus, the corrected sequence is:

Word	Initial Tag	Corrected Tag	Reason
economic	JJ	JJ	Describes the noun “growth.”
growth	NN	NN	Singular noun and subject of the sentence.
increases	NNS	VBZ	Third-person singular present-tense verb.
employment	NN	NN	Noun functioning as the object of “increases.”

---

2. Analysis of the Initial Tagging Errors

The main initial tagging error is:
increases/NNS → increases/VBZ
Why NNS is incorrect
NNS means *plural common noun*. If “increases” were an NNS, it would normally represent plural instances of an increase, as in:
“The company reported several increases.”
Here, however, “increases” is performing an action. It tells us what economic growth does to employment.
Why VBZ is correct
VBZ represents a third-person singular present-tense verb.
The subject is:

Economic growth
The head of this noun phrase is growth, which is singular. Therefore, the appropriate verb is:
growth increases
rather than:
growth increase

The grammatical structure is:
where:

- Economic = adjective (JJ)
- growth = singular noun (NN)
- increases = singular present verb (VBZ)
- employment = noun (NN)

Semantic justification
The sentence describes a cause-and-effect relationship:
Economic growth increases employment.

Therefore, *increases* must express an action or relationship between economic growth and employment, making VBZ the appropriate tag.

---

3. Word Frequency Distribution

The corpus contains:

Word	Frequency
economic	120
growth	450
increases	210
employment	380

Total frequency

Therefore, the total number of word occurrences is 1,160.

The frequency distribution is:

Calculations

economic

≈ 10.34%

growth

≈ 38.79%

increases

≈ 18.10%

employment

≈ 32.76%

Frequency distribution table

Word	Frequency	Probability	Percentage
economic	120	0.1034	10.34%
growth	450	0.3879	38.79%
increases	210	0.1810	18.10%
employment	380	0.3276	32.76%
Total	1160	1.0000	100%

How frequency supports probabilistic POS tagging

Corpus frequency provides statistical evidence about how words are normally used. A POS tagger can estimate probabilities such as:

For example, if "increases" occurs predominantly as a verb in a large tagged corpus, the system can assign a high probability to:

Similarly, contextual probabilities can be considered:

This is particularly useful for ambiguous words, where the same word can have multiple POS tags.

However, word frequency alone cannot determine the POS tag. Context and neighboring tags are also important. The Brill Tagger uses learned transformation rules to exploit such contextual information.

---

4. Transformation-Based Tagging and Entropy

Transformation-Based Tagging

A Transformation-Based Tagger, such as the Brill Tagger, typically works in two stages:

1. An initial tagger assigns POS tags.
2. Learned transformation rules identify common errors and correct them.

In this case:
Before transformation
economic/JJ growth/NN increases/NNS employment/NN
Transformation rule
Change NNS → VBZ if the preceding word is tagged NN.
Since growth/NN precedes increases/NNS, the rule applies.
After transformation
economic/JJ growth/NN increases/VBZ employment/NN
This improves the tagging accuracy because the transformation incorporates contextual grammatical information that the initial tagger missed.

Advantages
Transformation-Based Tagging:

- Corrects systematic tagging errors.
- Uses linguistic context.
- Learns rules from training data.
- Is relatively interpretable because each correction can be explained by a rule.
- Can improve the output of a simpler initial tagger.
- Is useful for processing large collections of text when common tagging errors can be identified systematically.

Overall Evaluation

The initial tagger incorrectly identifies “increases” as NNS, but the Brill transformation recognizes the contextual pattern NN + NNS and changes the tag to VBZ. This is grammatically and semantically correct because “economic growth” is a singular subject and “increases” is its present-tense verb.
Corpus frequencies provide useful statistical evidence for probabilistic tagging, while transformation rules provide context-sensitive corrections. Entropy can then be used to quantify the uncertainty of the tagger: lower entropy after a successful transformation indicates greater confidence in the corrected POS assignment.
Final Corrected Sentence
economic/JJ growth/NN increases/VBZ employment/NN

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________